Financial factors, firm size and potential

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Financial frictions theory: Overview

- **Theory:** Credit market frictions can endogenously amplify business cycles
- **Key prediction:** Differences in cyclical behavior should emerge across firms, depending on their financial position
- **Common proxy:** Size
  Financial frictions are arguably stronger for small firms since they tend to:
  - be younger
  - have more idiosyncratic risk
  - rely heavily on intermediary credit
  - face liquidity constraints
  - etc.

Two papers, two conclusions

Gertler and Gilchrist (1994)

- Use size as a proxy for financial constraints in the QFR aggregate data
- Small firms are about twice as cyclical compared to large ones after a Romer (1992) tight money date
- This can have aggregate effects as small firms account for about 60% of the sales decline

Crouzet and Mehrotra (2020)

- Only the largest 1% of firms in the QFR micro data are less cyclical
- This heterogeneity does not matter in the aggregate
- Suggest that market access is responsible for size effect

Our contribution

Empirical

1. Financial factors - when measured more directly - do matter for firm cyclicality
2. This effect is orthogonal to the size effect documented in the literature
3. Financially constrained firms exist in every size bin

Theoretical (still work in progress)

1. Develop a model with financial frictions where firms differ both ex-ante and ex-post
2. Size distribution features financially constrained firms almost everywhere
3. Amplified aggregate effects due to large constrained firms

Structure of the talk

1. Literature
2. Stylized facts
3. Statistical model
4. Structural model
5. Model predictions

Literature

Literature

1. **Financial frictions on the firm side:**
   Gertler & Gilchrist (1994) , Kashyap et al. (1994),Cooley & Quadrini (2001), Khan & Thomas (2013), Ottonello & Winberry (2018), Mehrotra & Sergeyev (2020),
2. **Size and cyclicality**
   Kudlyak & Sanchez (2017), Crouzet & Mehrotra (2020), Pugsley et al. (2021),
3. **Measuring financial constraints**
   Kaplan & Zingales (1997), Lamont & Polk (2001), Whited & Wu (2006), Farre-Mensa & Ljungqvist (2016)

Stylized facts

Data

- Use *Informação Empresarial Simplificada (IES)* data on the universe of Portuguese firms between 2006 and 2017
- Matched with Bank of Portugal credit register that records individual bank relationships and respective credit situations
- This data set is ideal for an analysis of corporate financial constraints:
  - Data on potential credit: Banks have to report potential credit of their customers (credit lines, credit cards, etc.)
  - Very detailed: Any (potential) loan amounting to 50 Euros or more is recorded in the credit register

Regression design

Run the following regressions, in the spirit of Crouzet & Mehrotra (2020):

$$g_{it} = \Delta GDP_t + \sum_{j \in \mathcal{J}} (\alpha_j + \beta_j \Delta GDP_t) 1_{i \in \mathcal{S}_t^{(j)}} + (\zeta + \eta \Delta GDP_t) fin_health_{it}$$
$$+ \sum_{l \in \mathcal{L}} (\gamma_l + \delta_l \Delta GDP_t) 1_{i \in \mathcal{L}} + \epsilon_{it}$$

- g_{it} is the year-on-year log change in turnover or employees
- the set $\mathcal{S}_t^{(j)}$ is a jth size group, e.g. all firms above the 90th but below the 99th percentile
- *fin_health* refers to the variable measuring the strength of financial constraints
- $\mathcal{L}$ is a set of industry dummies

Financial health variables

We define a battery of financial health variables:

- Constrained $= 1_{\text{Potential Credit}_t = 0}$
- Constrained Adj $= 1_{\text{Potential Credit}_t = 0 \ \& \ \text{Potential Credit}_{t-1} > 0}$
- Constrained Adj Eff $= 1_{\text{Potential Credit}_t = 0 \ \& \ \Delta\text{Effective credit} < 0}$
- Constrained Due $= 1_{\text{Potential Credit}_t = 0 \ \& \ \Delta\text{Overdue Credit}_t > 0}$

Cyclicality of Turnover

	Turnover growth				
	(1)	(2)	(3)	(4)	(5)
GDP growth	2.505***	2.428***	2.443***	2.470***	2.454***
	(0.023)	(0.026)	(0.023)	(0.028)	(0.025)
[90,99] × GDP growth	0.035	0.068	0.063	0.075	0.056
	(0.107)	(0.107)	(0.107)	(0.121)	(0.121)
[99,99.5] × GDP growth	-0.689**	-0.656**	-0.660**	-0.742**	-0.754**
	(0.300)	(0.301)	(0.300)	(0.342)	(0.341)
[99.5,100] × GDP growth	-1.469***	-1.426***	-1.434***	-1.607***	-1.646***
	(0.293)	(0.293)	(0.293)	(0.359)	(0.359)
Constrained × GDP growth		0.119**			
		(0.055)			
Constrained adj × GDP growth			0.572***		
			(0.135)		
Constrained adj eff × GDP growth				0.165**	
				(0.069)	
Constrained due × GDP growth					1.202***
					(0.251)
Observations	1,323,660	1,323,660	1,323,660	1,161,187	1,159,037
R-squared	0.029	0.030	0.030	0.030	0.032
Industry FE	Yes	Yes	Yes	Yes	Yes
Industry FE × GDP growth	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Sales Employment

Discussion of results

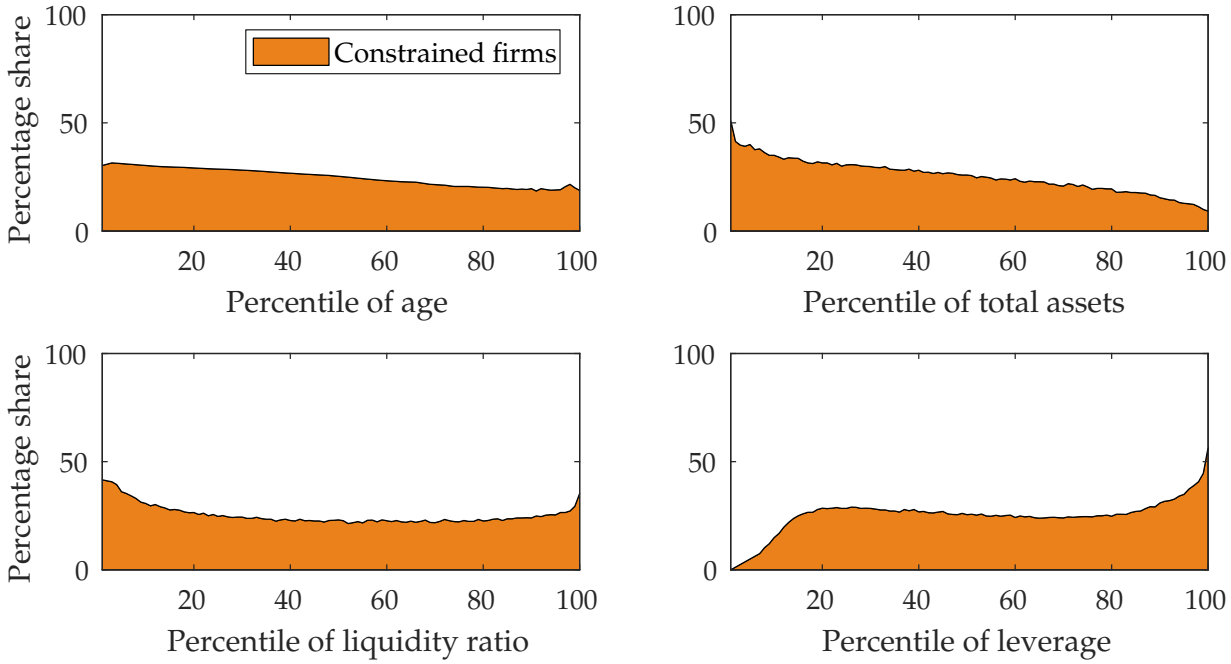
We find that:

- Financial frictions matter for firm cyclicality
- The interaction coefficient of constrained firms and GDP growth is positive and significant
- Confirmation of a broader financial accelerator story

and, simultaneously, that

- Size matters also for firm cyclicality, just like in Crouzet and Mehrotra (2020)
- But this seems to be orthogonal to the financial health!
- Confirmation of a non-financial story, too

Proxies for financial constraints



- There are constrained firms in every bin
- The correlation between proxies and constraints is as we would expect
- But this correlation is not very strong!
  E.g. a two standard deviation increase in assets (going from the 5th to the 95th percentile) decreases the probability of being constrained only by ten percent

Discussion

- Standard financial frictions models predict very strong correlation between size and being financially constrained
- Could ex-ante conditions break this correlation?
  - Small firms may be unconstrained as they already reached their potential...
  - ...while large firms may still be growing and may still be constrained
- Different ex-ante potential would create a dispersion of constrained firms across the entire firm size distribution

  sketch
- But is it reasonable to assume ex-ante heterogeneity?
  $\rightarrow$ Some recent affirmative evidence by Pugsley et al. (2021)

Statistical model

Statistical model

In order to get a first understanding of ex-ante and ex-post components we estimate the Pugsley et al. (2021) model on both constrained and unconstrained firms

$$\underbrace{\ln n_{i,a}}_{\text{log employment}} = \underbrace{u_{i,a} + v_{i,a}}_{\text{Ex-ante component}} + \underbrace{w_{i,a} + z_{i,a}}_{\text{Ex-post component}}$$

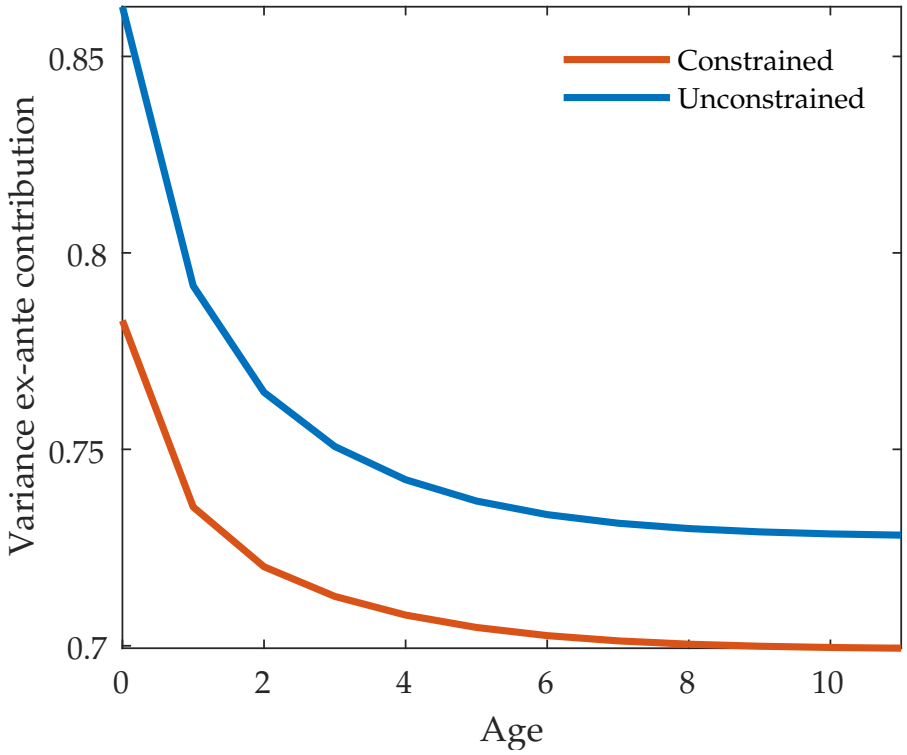
where

$$\begin{aligned}
u_{i,a} &= \rho_u u_{i,a-1} + \theta_i, & u_{i,-1} &\sim iid\left(\mu_{\tilde{u}}, \sigma^2_{\tilde{u}}\right), & \theta_i &\sim iid\left(\mu_\theta, \sigma^2_\theta\right), & |\rho_u| &\leq 1 \\
v_{i,a} &= \rho_v v_{i,a-1}, & v_{i,-1} &\sim iid\left(\mu_{\tilde{v}}, \sigma^2_{\tilde{v}}\right), & & & |\rho_v| &\leq 1 \\
w_{i,a} &= \rho_w w_{i,a-1} + \varepsilon_{i,a}, & w_{i,-1} &= 0, & \varepsilon_{i,a} &\sim iid\left(0, \sigma^2_\varepsilon\right), & |\rho_w| &\leq 1 \\
z_{i,a} &\sim iid\left(0, \sigma^2_z\right)
\end{aligned}$$

Estimation details Figure: model fit

Ex-ante component variance contribution

Can calculate the contribution of the ex-ante component to the total variance at different ages:



Ex-ante contribution expression

Autocovariance expression

Structural model

Structural model - overview

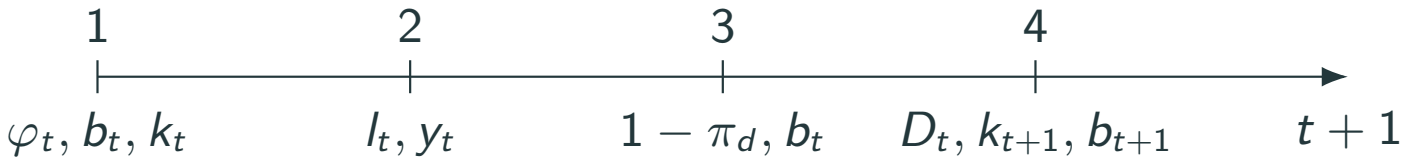
- Representative household, chooses labour and consumption details
- Firms produce according to the following production function:

$$y = \underbrace{A}_{MITshocked} \varphi k^{\alpha} l^{\nu}$$

- Idiosyncratic productivity schedule with transitory and permanent components. details
- Firms maximize profits subject to a collateral constraint
- Standard entry assumptions details

Incumbents - Within-period timing

The within-period timing of an incumbent firm i can be illustrated as follows (dropping respective subscript below):



1. Observes its idiosyncratic productivity φ_t, current stock of debt b_t and capital k_t
2. Chooses labour input and production details
3. If hit by the stochastic, exogenous death shock d it repays outstanding debt b_t and exit
4. Conditional on survival, firm chooses its investment k_{t+1} and borrowing b_{t+1} subject to the borrowing constraint $b_{t+1} \leq (\xi A_t^{\iota}) k_{t+1}$

Firm problem after production

- Expected value of the firm after production is:

$$V^1(x_t, \varphi_t) = \pi_d x_t + (1 - \pi_d)\, V^2(x_t, \varphi_t)$$

where $x_t \equiv \pi(k_t, \varphi_t) + (1 - \delta)k_t - b_t$ is current cash-on-hand

- Surviving firms face the following optimisation problem:

$$V^2(x_t, \varphi_t) = \max_{k_{t+1}, b_{t+1}} \left\{ D_t + \mathbb{E}_t \left[\Lambda_{t,t+1} V^1 \left(x_{t+1}, \varphi_{t+1} \right) | \varphi_t \right] \right\}$$

s.t.

$$D_t \equiv x_t + q_t b_{t+1} - k_{t+1} \geq 0$$

$$b_{t+1} \leq (\xi A_t^{\iota}) k_{t+1}$$

Who are the constrained?

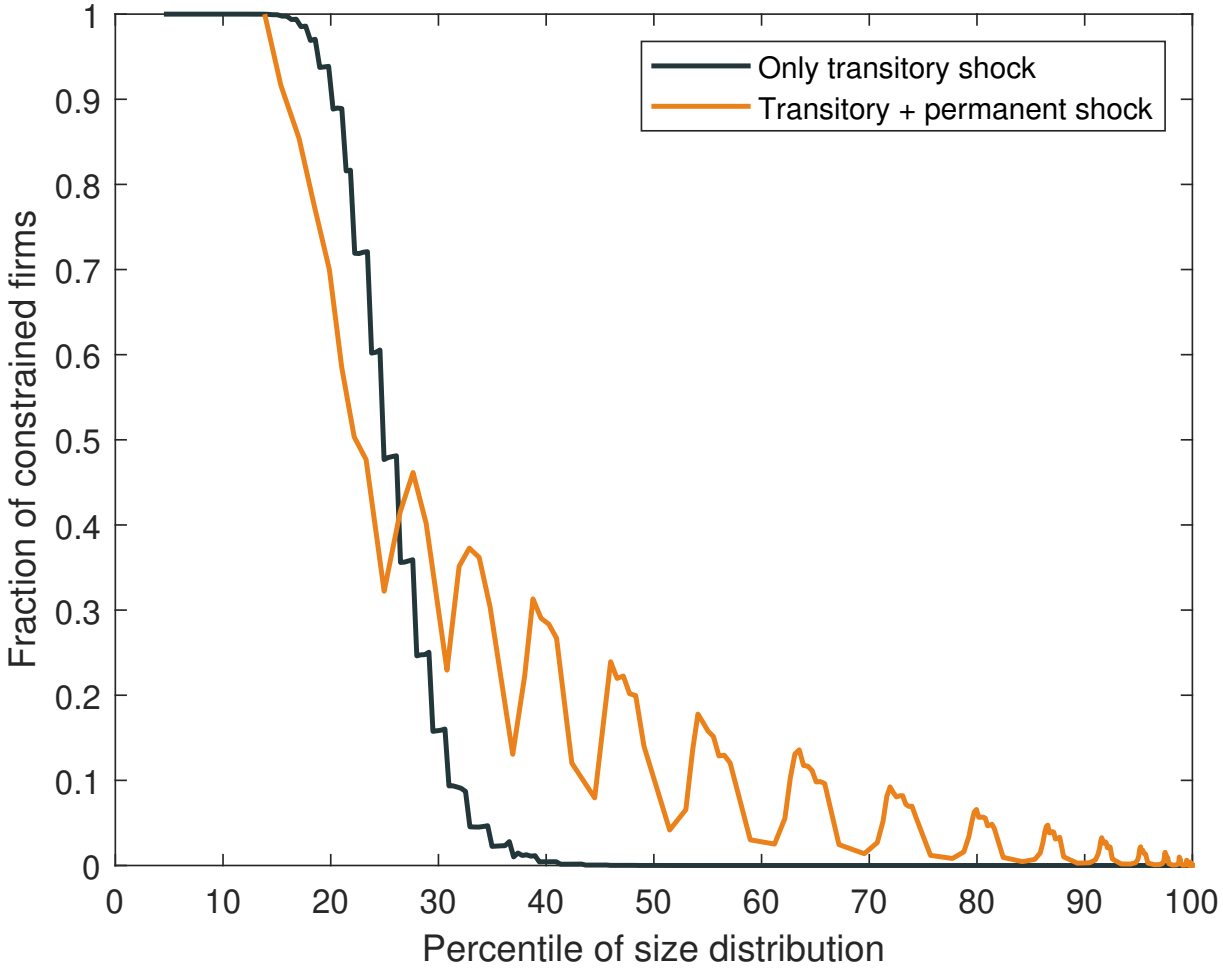
The **optimal, unconstrained amount** of capital $k^*(\varphi_{it})$ solves the following equation

$$\beta \mathbb{E}_t \left[\frac{d\pi(\varphi_{it+1}, k_{it+1})}{dk_{it+1}} | \varphi_{it} \right] + \beta(1-\delta) = 1$$

We consider all firms that cannot implement $k^*(\varphi_{it})$ as constrained.

Model predictions

Steady state distributions - constrained and unconstrained

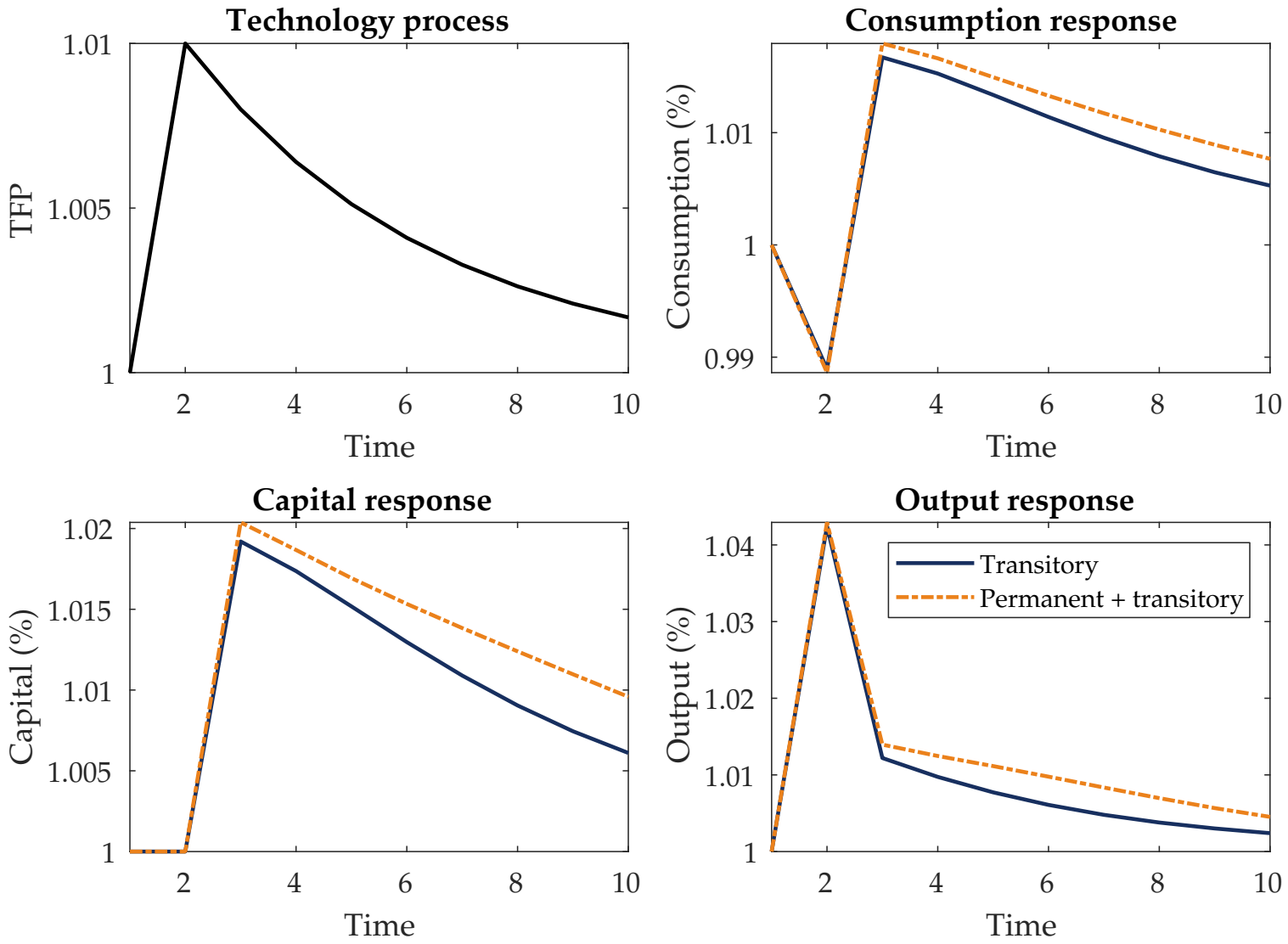


Calibration

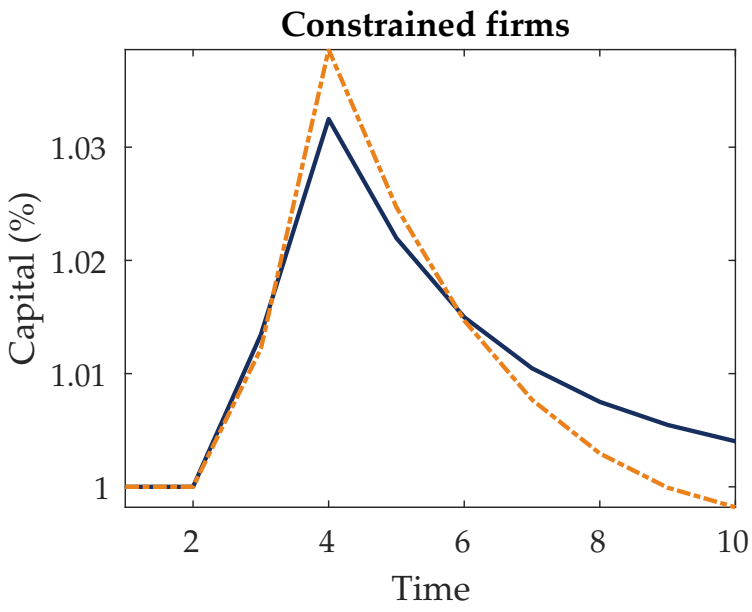
Moment	Data	Model trans. + perm. shock	Model transitory shock
Targeted moments			
Percentage of constrained firms	0.23	0.21	0.22
Size of 90th-percentile vs. median	9.44	9.32	11.09
Size of constrained firms 90th-percentile vs. median	7.35	7.75	8.34
Size of unconstrained firms 90th-percentile vs. median	9.67	9.25	8.65
Capital share of constrained firms	0.12	0.17	0.12
Untargeted moments			
Share of constrained firms in top 10% vs. bottom 20%	0.36	0.23	0
Percentage of constrained firms in top 1%	0.09	0.09	0

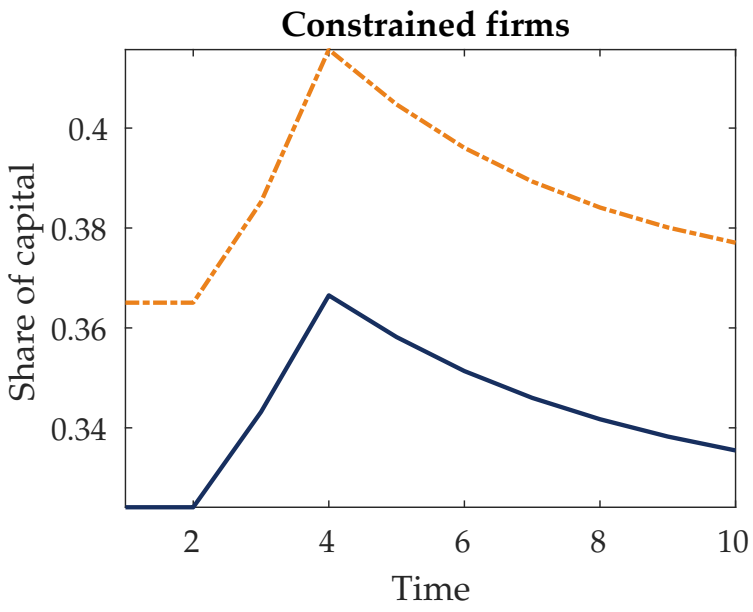
Parameter details

Aggregate response



Mechanisms at play





Conclusion

- **Empirical**
  - Financial frictions matter for firm cyclicality.
  - Size effects orthogonal to financial frictions.
- **Model**
  - Introduction of permanent shock allows us to match the distribution of constrained
  - Larger share of capital in constrained firms
  - Stronger aggregate response due to large constrained firms
  - But: Still hinges on cyclicality of collateral constraint $\rightarrow$ endogenize.

Appendix

C&M: Do small and large firms react heterogeneously to aggregate shocks?

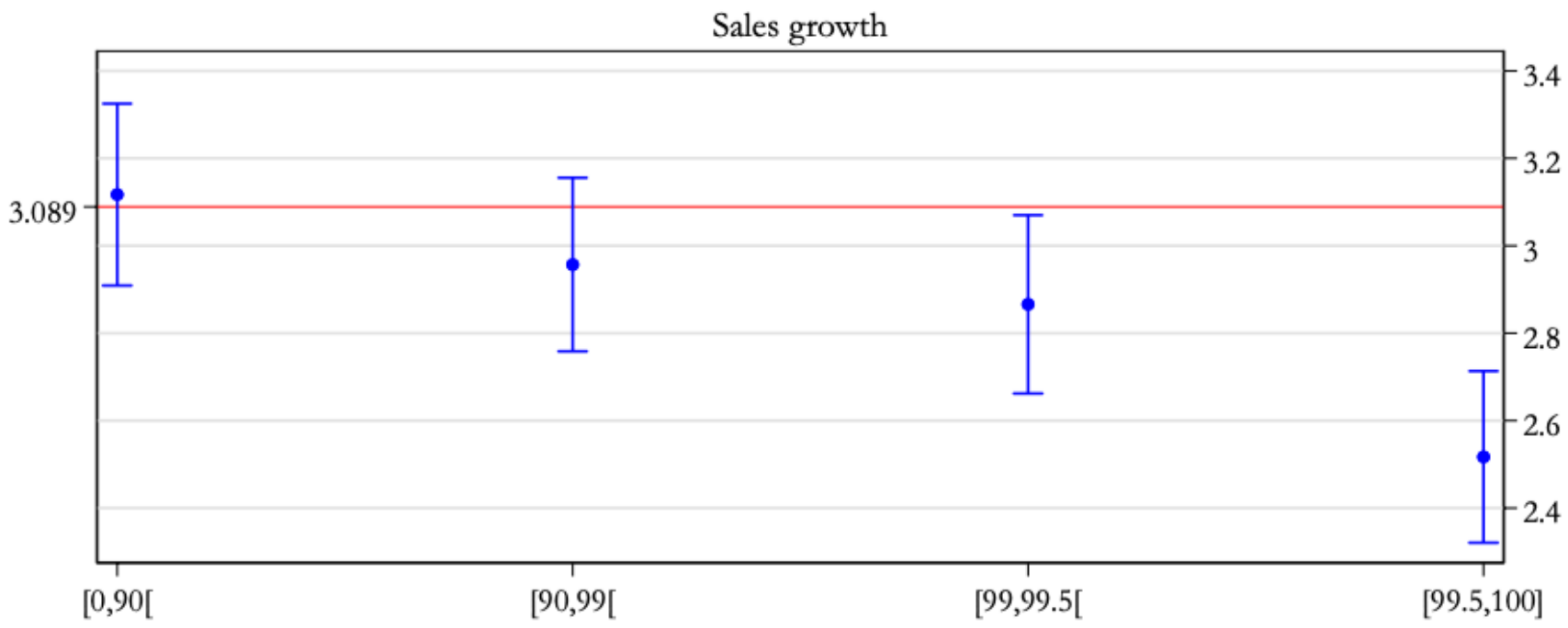
Yes! But only at the very top.

	Sales growth	Inventory growth	Fixed investment
$[90, 99] \times$ GDP growth	-0.160	-0.107	-0.299^{*}
	(0.142)	(0.174)	(0.157)
$[99, 99.5] \times$ GDP growth	-0.251^{*}	-0.299^{*}	-0.687^{***}
	(0.143)	(0.180)	(0.194)
$[99.5, 100] \times$ GDP growth	-0.600^{***}	-0.730^{***}	-1.257^{***}
	(0.140)	(0.206)	(0.355)
Observations	≈ 460000	≈ 460000	≈ 460000
Firms	≈ 60000	≈ 60000	≈ 60000
Adj. R^2	0.025	0.006	0.003
Clustering	Firm	Firm	Firm
Industry controls	D/ND	D/ND	D/ND

Figure for sales growth

Figure for inventory growth and investment

C&M: Average marginal effects of GDP growth



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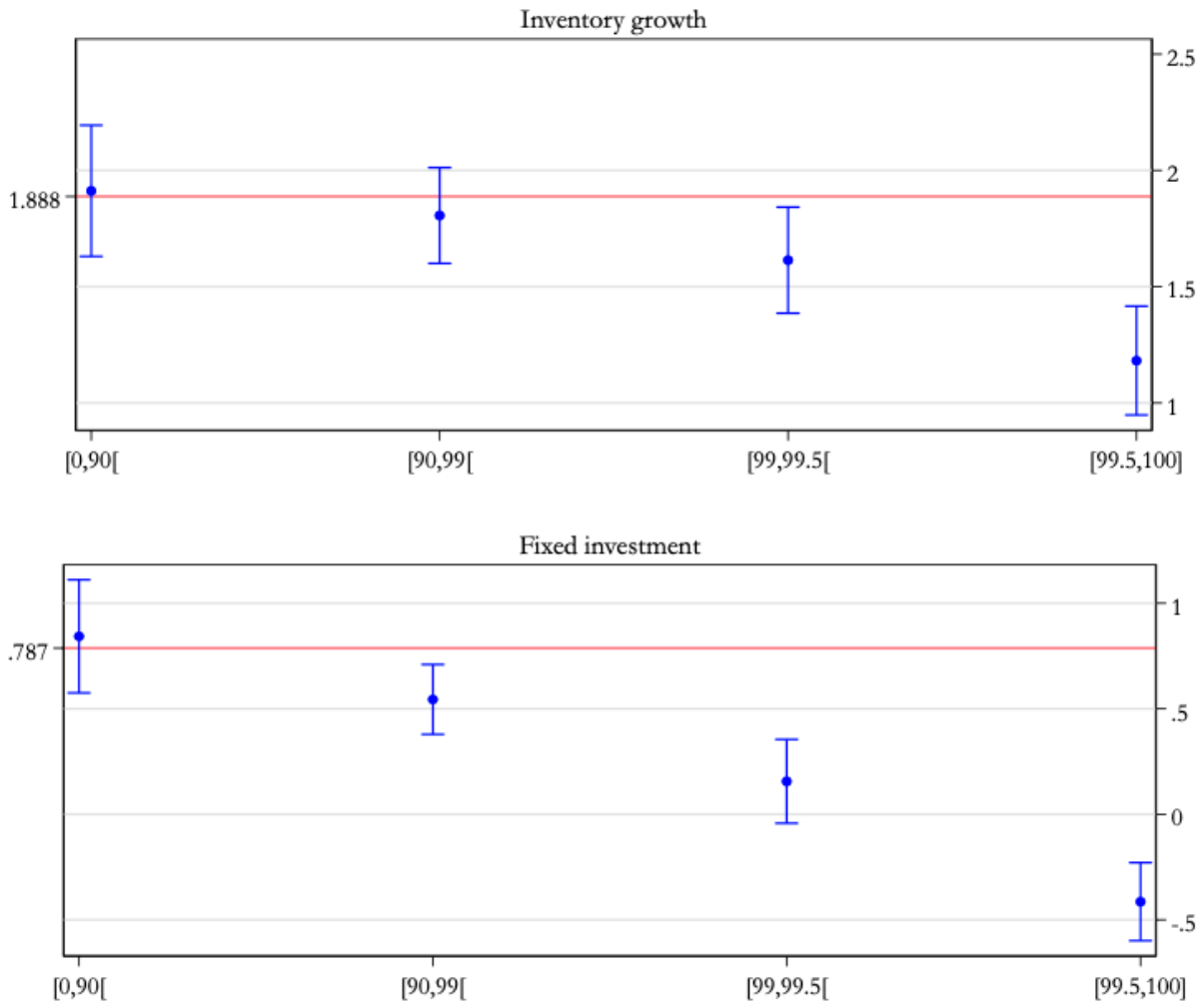
C&M: Average marginal effects of GDP growth



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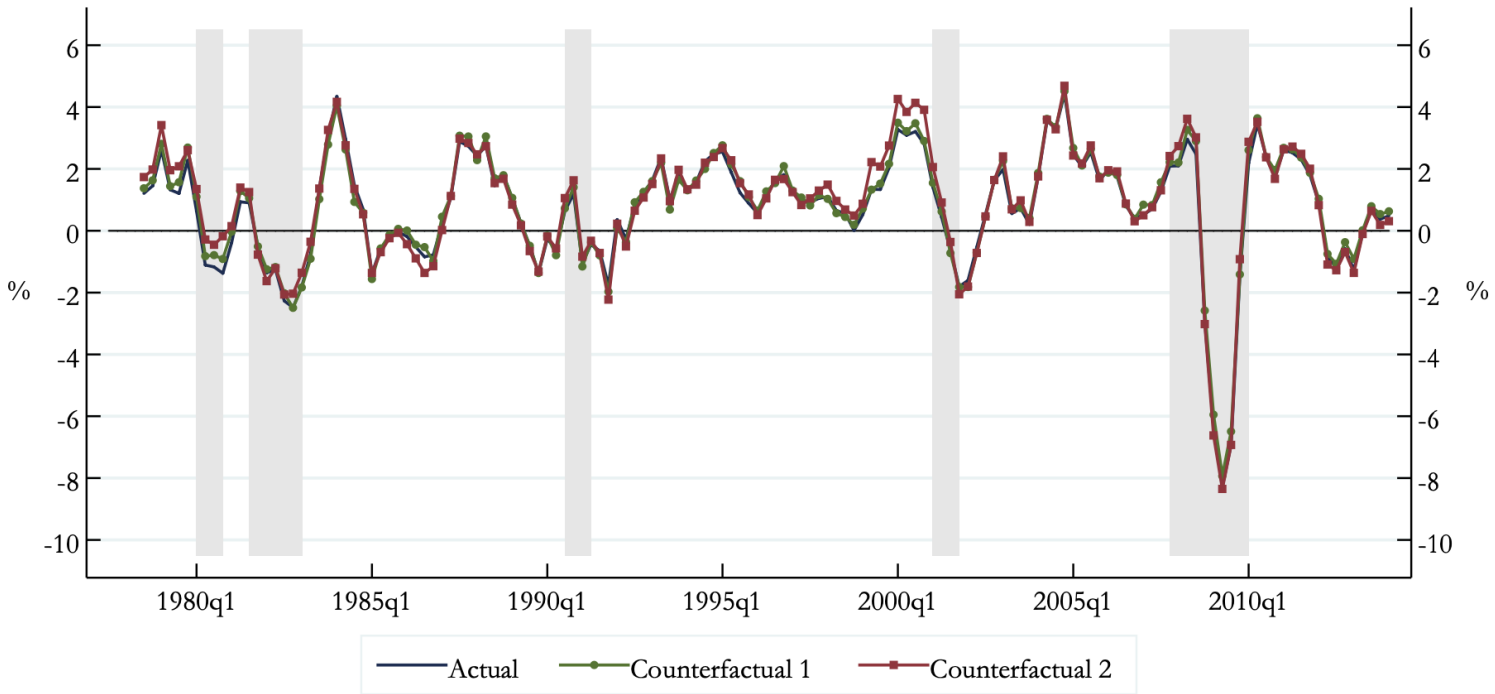
C&M: Counterfactual sales growth series



- Counterfactual growth rate G(1) (circled green line) assumes that the average firm-level growth rate of small and large firms is equal
- Counterfactual growth rate 2 G(2) (squared red line) also assumes that the covariance between size and growth is the same

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Cyclicality of Employment

	Employees growth						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
[90,99] × GDP growth	0.003	0.007	0.014	-0.107**	0.001	-0.001	-0.110**
	(0.046)	(0.047)	(0.046)	(0.052)	(0.047)	(0.046)	(0.052)
[99,99.5] × GDP growth	-0.003	-0.003	0.006	-0.154	0.005	-0.011	-0.151
	(0.130)	(0.131)	(0.130)	(0.152)	(0.130)	(0.130)	(0.152)
[99.5,100] × GDP growth	-0.328**	-0.327**	-0.316**	-0.395**	-0.312**	-0.339**	-0.411**
	(0.150)	(0.150)	(0.150)	(0.176)	(0.150)	(0.150)	(0.176)
Constrained × GDP growth		-0.021					
		(0.029)					
Constrained Adj × GDP growth			0.131*				
			(0.067)				
Constrained Due× GDP growth				0.318***			
				(0.121)			
Const + potential const × GDP growth					0.066**		
					(0.026)		
Constrained continuous × GDP growth						-0.80***	
						(0.013)	
Constrained continuous 2 × GDP growth							-0.088***
							(0.016)
Observations	1,287,296	1,287,296	1,287,296	1,126,902	1,287,296	1,286,004	1,125,990
R-squared	0.024	0.024	0.025	0.026	0.025	0.024	0.025
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE × GDP growth	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

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Cyclicality of Sales

	Sales growth						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
[90,99] × GDP growth	0.220	0.267*	0.247*	0.308*	0.238	0.211	0.296*
	(0.150)	(0.151)	(0.150)	(0.170)	(0.151)	(0.150)	(0.170)
[99,99.5] × GDP growth	-1.012**	-0.968**	-0.986**	-1.419***	-0.985**	-1.023**	-1.407***
	(0.446)	(0.446)	(0.446)	(0.526)	(0.446)	(0.446)	(0.526)
[99.5,100] × GDP growth	-1.708***	-1.650***	-1.674***	-1.775***	-1.678***	-1.725***	-1.824***
	(0.472)	(0.473)	(0.472)	(0.570)	(0.472)	(0.472)	(0.571)
Constrained × GDP growth		0.186*					
		(0.098)					
Constrained Adj × GDP growth			0.296				
			(0.222)				
Constrained Due× GDP growth				0.936**			
				(0.432)			
Const + potential const × GDP growth					0.111		
					(0.086)		
Constrained continuous × GDP growth						-0.155***	
						(0.056)	
Constrained continuous 2 × GDP growth							-0.186***
							(0.061)
Observations	803,742	803,742	803,742	708,881	803,742	803,326	708,598
R-squared	0.023	0.024	0.024	0.025	0.023	0.023	0.024
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE × GDP growth	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

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Linear probability model

	Constrained binary			
	(1)	(2)	(3)	(4)
Age	-0.05***			
	(0.000)			
Total assets		-0.05***		
		(0.001)		
Leverage			0.03***	
			(0.000)	
Liquidity ratio				0.01***
				(0.000)
Constant	0.36***	0.36***	0.36***	0.36***
Observations	1,765,288	1,765,288	1,764,947	1,764,947
R-squared	0.011	0.000	0.000	0.000

Notes: Standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

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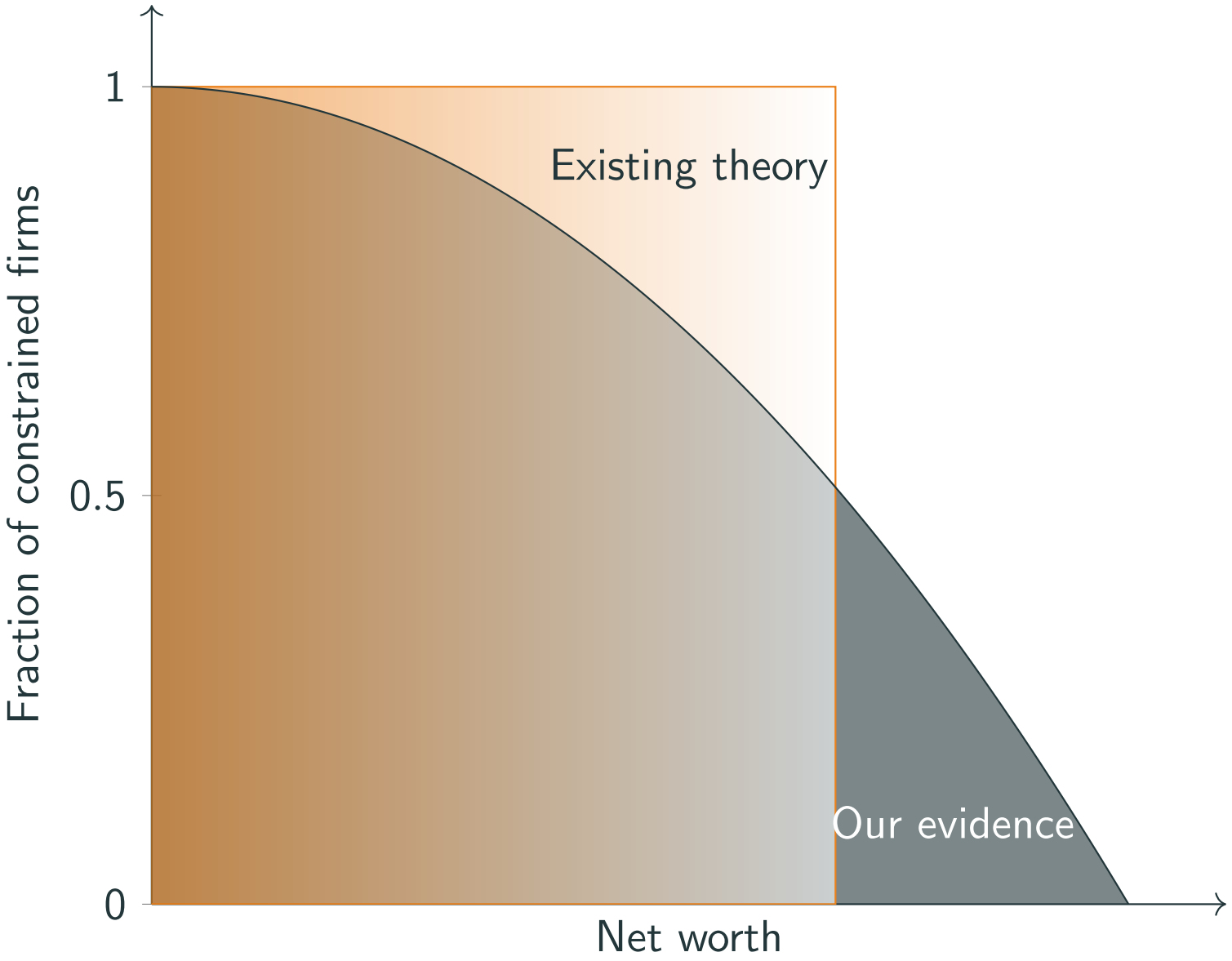
Evidence v Theory



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Standard deviation and autocorrelation of employment

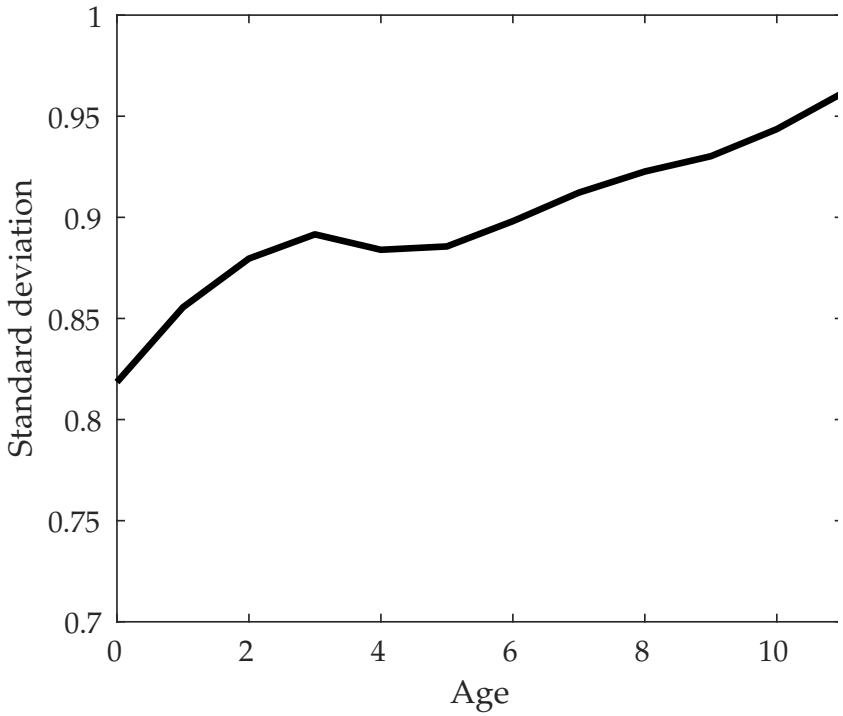

Figure 1: Standard deviation

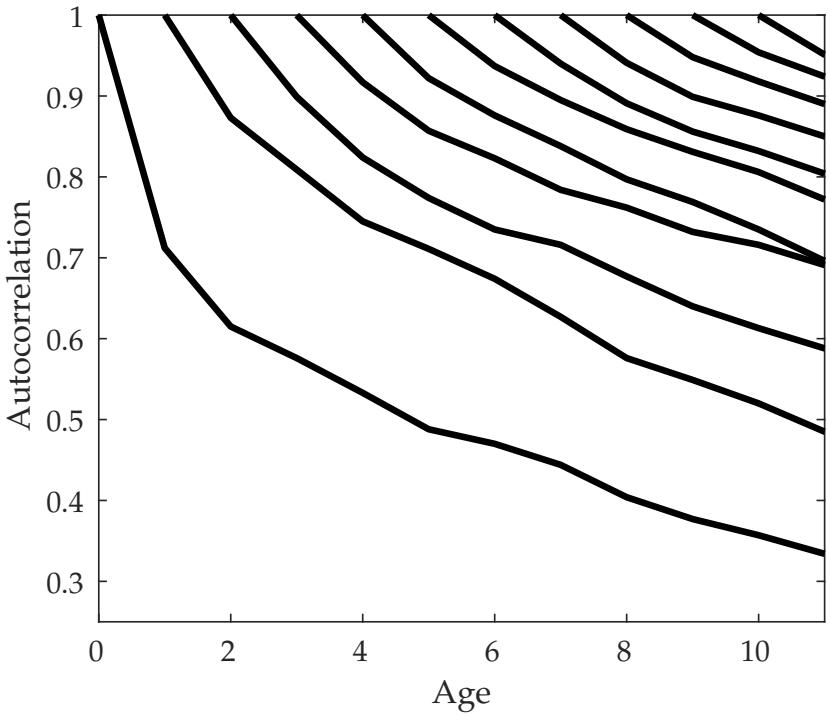

Figure 2: Autocorrelation

Constrained vs unconstrained
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Statistical model

In order to get a first understanding of ex-ante and ex-post components we estimate the Pugsley et al. (2020) model on both constrained and unconstrained firms

$$\underbrace{\ln n_{i,a}}_{\text{log employment}} = \underbrace{u_{i,a} + v_{i,a}}_{\text{Ex-ante component}} + \underbrace{w_{i,a} + z_{i,a}}_{\text{Ex-post component}}$$

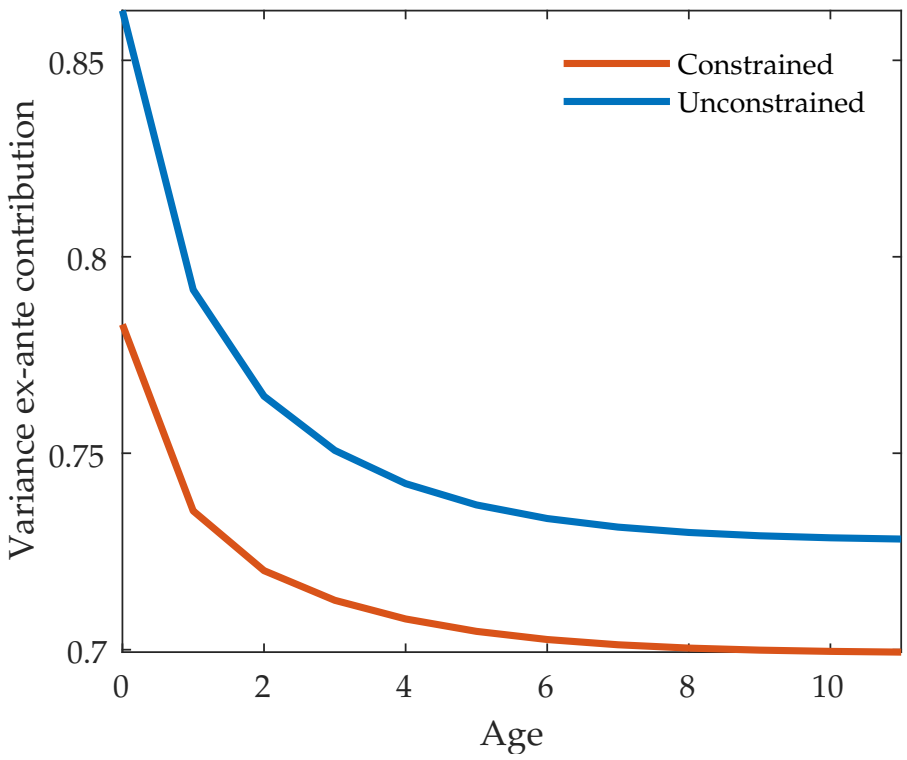
where

$$\begin{aligned}
u_{i,a} &= \rho_u u_{i,a-1} + \theta_i, & u_{i,-1} &\sim iid\left(\mu_{\tilde{u}}, \sigma^2_{\tilde{u}}\right), & \theta_i &\sim iid\left(\mu_\theta, \sigma^2_\theta\right), & |\rho_u| &\leq 1 \\
v_{i,a} &= \rho_v v_{i,a-1}, & v_{i,-1} &\sim iid\left(\mu_{\tilde{v}}, \sigma^2_{\tilde{v}}\right), & & & |\rho_v| &\leq 1 \\
w_{i,a} &= \rho_w w_{i,a-1} + \varepsilon_{i,a}, & w_{i,-1} &= 0, & \varepsilon_{i,a} &\sim iid\left(0, \sigma^2_\varepsilon\right), & |\rho_w| &\leq 1 \\
z_{i,a} &\sim iid\left(0, \sigma^2_z\right)
\end{aligned}$$

Estimation details Figure: model fit back

Ex-ante component variance contribution

Can calculate the contribution of the ex-ante component to the total variance at different ages:



Ex-ante contribution expression

Autocovariance expression back

Standard deviation and autocorrelation by age

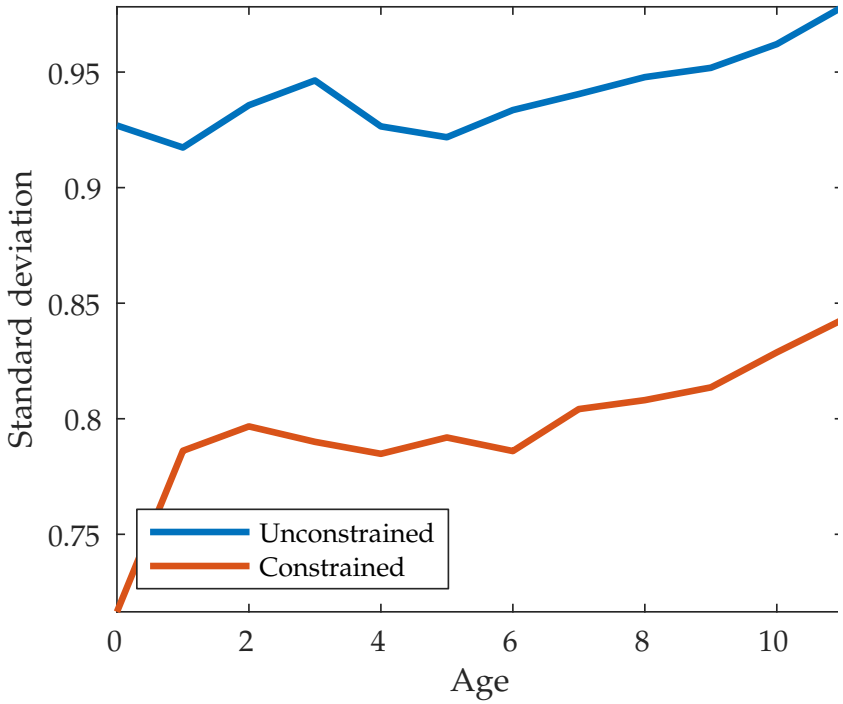


Figure 3: Standard deviation

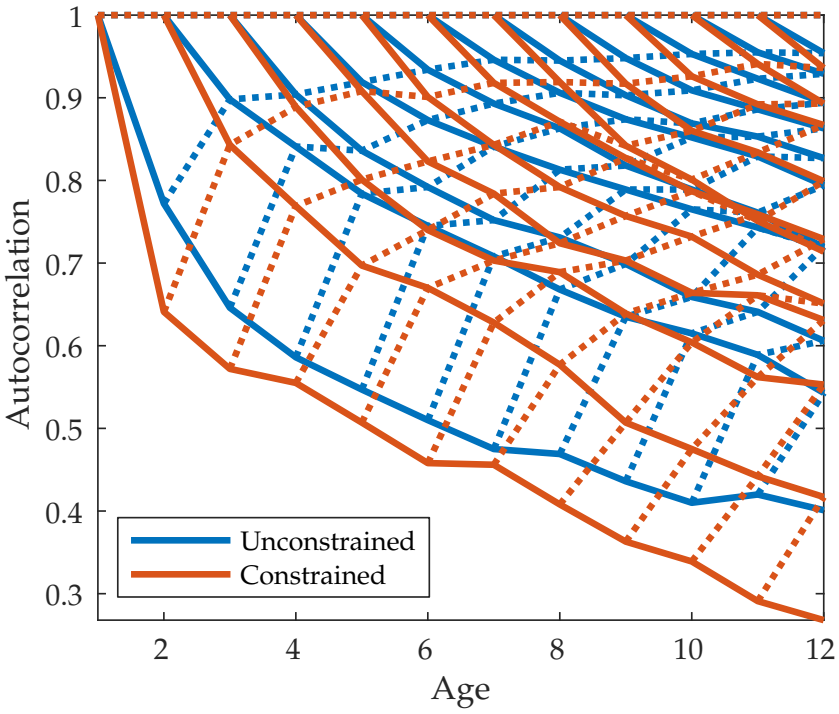


Figure 4: Autocorrelation

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Autocovariance

- Use autocovariance to estimate ex-ante and ex-post conditions importance
- Estimate autocovariance for two groups of firms:
  - Constrained: When a firm has potential credit equal to zero at age a-j
  - Unconstrained: Firms that have potential credit available at age a-j

$$Cov[\ln n_{i,a}, \ln n_{i,a-j}] = \underbrace{\left(\sum_{k=0}^{a} \rho_u^k\right)\left(\sum_{k=0}^{a-j} \rho_u^k\right)\sigma_\theta^2 + \rho_u^{2(a+1)-j}\sigma_{\hat{u}}^2 + \rho_v^{2(a+1)-j}\sigma_{\hat{v}}^2}_{\text{Ex-ante component}}$$

$$+ \underbrace{\sigma_\epsilon^2 \rho_w^j \sum_{k=0}^{a-j} \rho_w^{2k} + \sigma_z^2 1_{j=0}}_{\text{Ex-post component}}$$

Back Derivation

Derivation of autocovariance formula (1/2)

Write stochastic processes in MA representation:

$$u_{i,t} = \rho_u^{t+1} u_{i,-1} + \sum_{k=0}^{a} \rho_u^k \theta_i$$

$$v_{i,a} = \rho_v^{a+1} v_{i,-1}$$

$$w_{i,a} = \sum_{k=0}^{a} \rho_w^k \varepsilon_{i,a-k} = \sum_{k=0}^{j-1} \rho^k \varepsilon_{i,a-k} + \rho_v^j \sum_{k=0}^{a-j} \rho_v^k \varepsilon_{i,a-j-k} \quad 0 \leq j \leq a$$

So the level of log employment of firm i at age a is:

$$\ln n_{i,a} = \rho_u^{a+1} u_{i,-1} + \sum^{a} \rho_u^k \theta_i + \rho_v^{a+1} v_{i,-1} + \sum_{i=1}^{j-1} \rho^k \varepsilon_{i,a-k} + \rho_v^j \sum_{i=1}^{a-j} \rho_v^k \varepsilon_{i,a-j-k} + z_{i,a}$$

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Derivation of autocovariance formula (2/2)

Then the autocovariance of log employment at age a and $a-j$ for $j \geq 0$ is:

$$\begin{aligned}
\mathrm{Cov}\left[\log n_{i,a}, \log n_{i,a-j}\right] &= \left(\sum_{k=0}^{a} \rho_u^k\right) \sigma_\theta^2 \left(\sum_{k=0}^{a-j} \rho_u^k\right) + \rho_u^{a+1} \sigma_{\tilde{u}}^2 \rho_u^{a-j+1} + \rho_v^{a+1} \sigma_{\tilde{v}}^2 \rho_v^{a-j+1} \\
&\quad + \mathrm{Cov}\left[\rho_v^j \sum_{k=0}^{a-j} \rho_v^k \varepsilon_{i,a-j-k}, \sum_{k=0}^{a-j} \rho_v^k \varepsilon_{i,a-j-k}\right] + 1_{\{j=0\}} \sigma_z^2 \\
&= \sigma_\theta^2 \left(\sum_{k=0}^{a} \rho_u^k\right) \left(\sum_{k=0}^{a-j} \rho_u^k\right) + \sigma_{\tilde{u}}^2 \rho_u^{2(a+1)-j} + \sigma_{\tilde{v}}^2 \rho_v^{2(a+1)-j} + \sigma_\varepsilon^2 \rho_w^j \sum_{k=0}^{a-j} \rho_w^{2k} + 1_{\{j=0\}} \sigma_z^2
\end{aligned}$$

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Ex-ante contribution

$$\frac{\text{Ex-ante variance}}{\text{Total variance}}(\log n_{i,a}) =$$

$$\frac{\left(\sum_{k=0}^{a} \rho_u^k\right)^2 \sigma_\theta^2 + \rho_u^{2(a+1)} \sigma_{\hat{u}}^2 + \rho_v^{2(a+1)} \sigma_{\hat{v}}^2}{\left(\sum_{k=0}^{a} \rho_u^k\right)^2 \sigma_\theta^2 + \rho_u^{2(a+1)} \sigma_{\hat{u}}^2 + \rho_v^{2(a+1)} \sigma_{\hat{v}}^2 + \sigma_\epsilon^2 \sum_{k=0}^{a} \rho_w^{2k} + \sigma_z^2}$$

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Model parameters

	ρ_u	ρ_v	ρ_w	σ_θ	σ_u	σ_v	σ_ϵ	σ_z
Constrained	0.485	0.760	0.707	0.339	0.009	0.671	0.287	0.147
Unconstrained	0.447	0.694	0.780	0.452	0.760	0.841	0.305	0.110

- We minimize the sum of squared deviations between the empirical autocovariance and the model autocovariance
- Model calibrated to log employment, after controlling for birth year and industry fixed effects
- Long-run steady state level of employment given by

$$\ln n_i^* = \frac{\theta_i}{1 - \rho_u}$$

- For same initial shock, constrained firms will reach a higher steady state employment due to higher ρ_u

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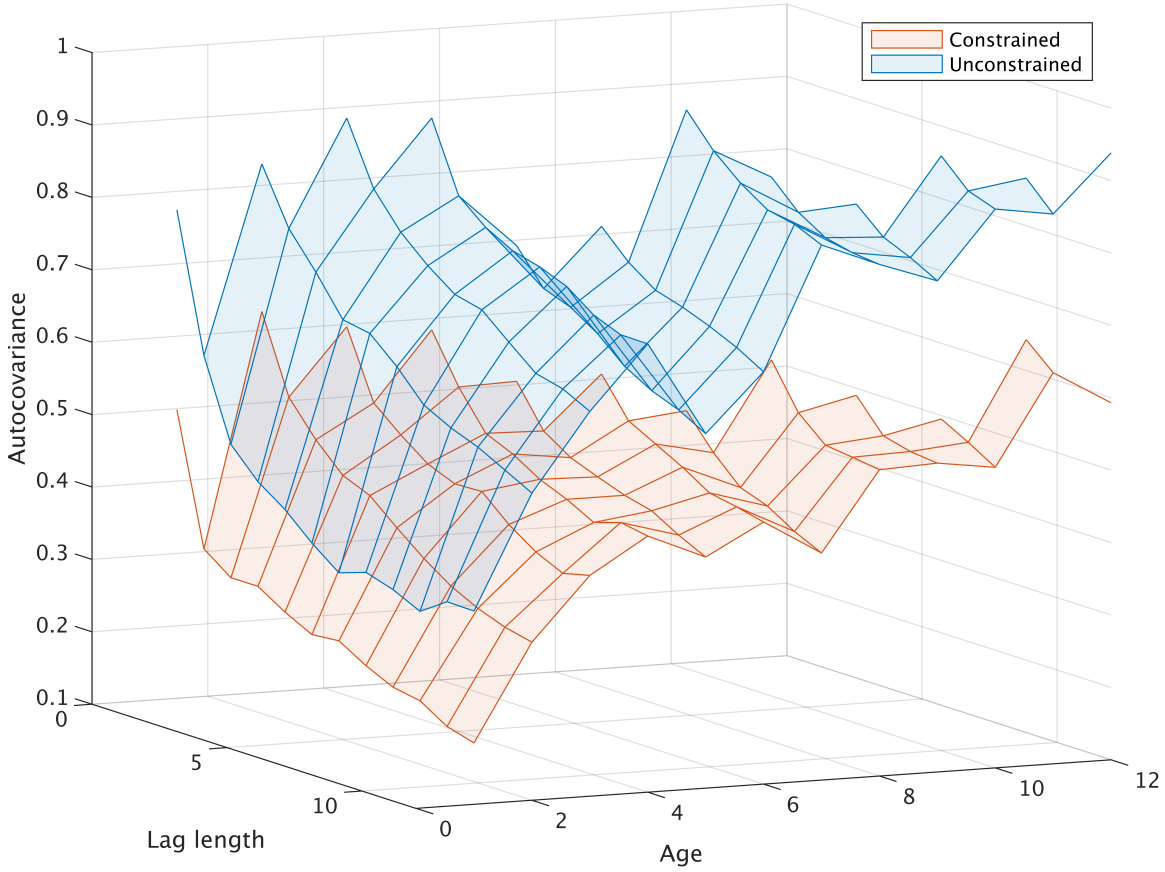
Autocovariance constrained vs unconstrained - data



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Autocovariance: model fit

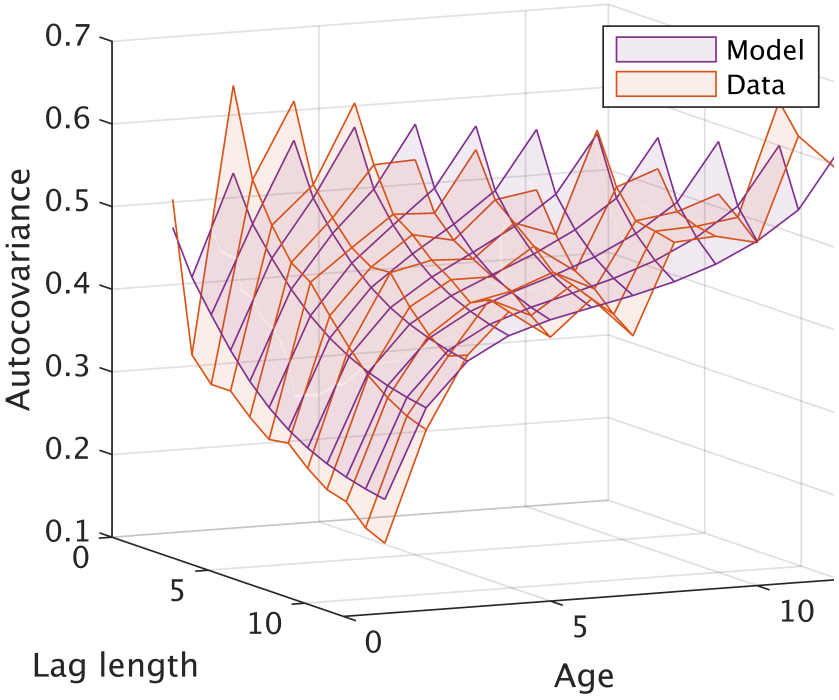

Figure 5: Constrained firms

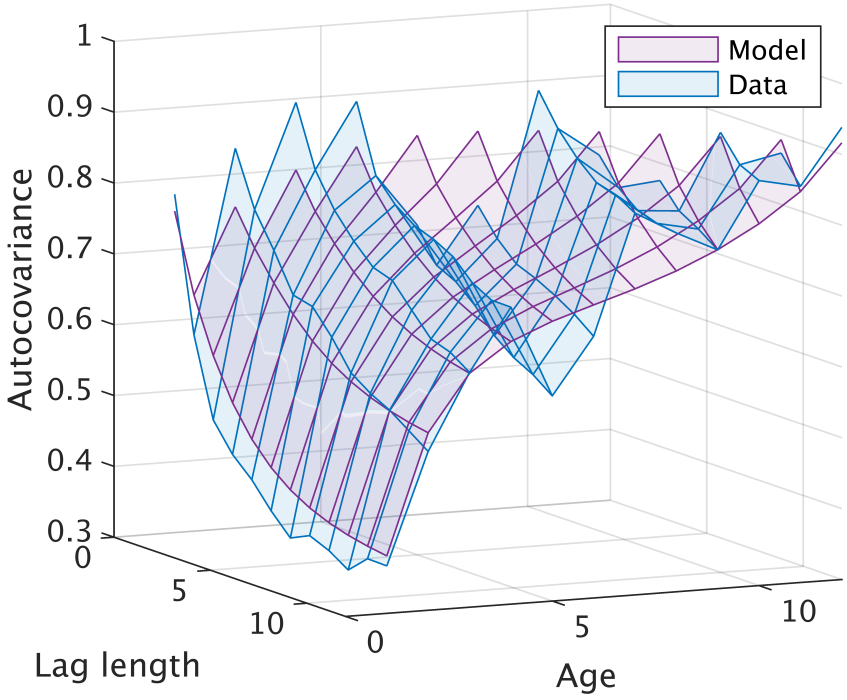

Figure 6: Unconstrained firms

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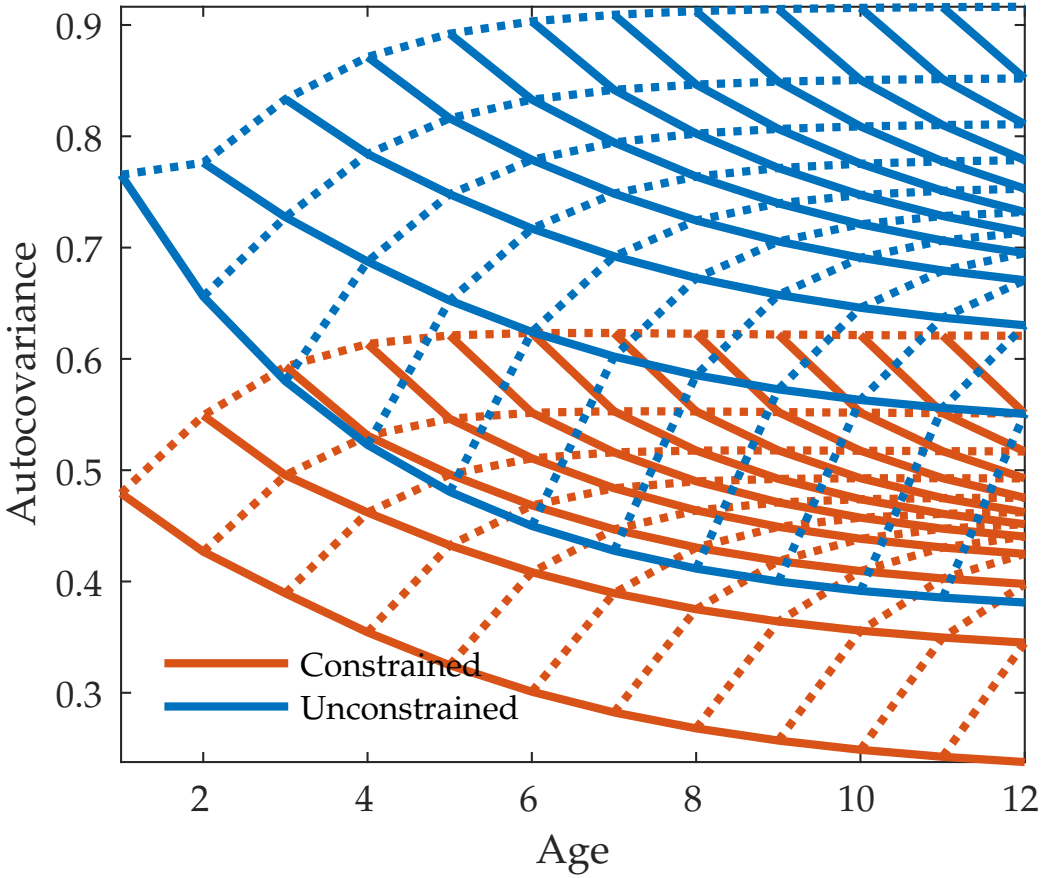
Autocovariance constrained vs unconstrained - model



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Households

A representative household solves the following recursive maximisation problem

$$V(k_t) = \max_{c_t, l_t, k_{t+1}} \{\log(c_t) - \chi l_t + \beta \mathbb{E}_t V(k_{t+1})\}$$

subject to:

$$c_t + k_{t+1} = (1 + r_t)k_{t+1} + \omega_t l_t + D_t$$

First order conditions are standard, pinning down interest rate and wages:

$$(1 + r_t) = \beta^{-1} \mathbb{E}_t \left[\frac{c_{t+1}}{c_t} \right]$$

$$\omega_t = \psi c_t$$

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Final goods producer

The final good producer buys products from intermediate goods producers and produces the final good Y_t using the following technology

$$Y_t = A_t \left(\int_0^1 y_{it}^{\left(\frac{\xi-1}{\xi}\right)} di \right)^{\left(\frac{\xi}{\xi-1}\right)}$$

The final goods firm maximizes profits by choosing the amount of intermediate goods from each supplier i:

$$\max_{y_{it}} \Pi_t = \max_{y_{it}} Y_t - \int_0^1 p_{it} y_{it} di$$

This yields the following first order condition:

$$p_{it} = A_t^{\frac{\xi-1}{\xi}} \left(\frac{Y_t}{y_{it}} \right)^{\frac{1}{\xi}} = A_t^{\frac{\xi-1}{\xi}} \left(\frac{Y_t}{\varphi_{it} k_{it}^{\alpha} l_{it}^{\upsilon}} \right)^{\frac{1}{\xi}}$$

Intermediate goods producer

Every intermediate goods producer faces the following static profit maximisation problem:

$$\begin{aligned}\max \pi_{it} =& p_{it} y_{it} - \omega_t l_{it} \\ =& Y_t^{1/\xi} \left(A_t \varphi_{it} k_{it}^{\alpha} l_{it}^{\upsilon}\right)^{\frac{\xi-1}{\xi}} - \omega_t l_{it}\end{aligned}$$

The first order condition with respect to l_{it} implies:

$$\frac{\omega_t}{\upsilon} \frac{\xi}{\xi - 1} Y_t^{-1/\xi} = \left(A_t \varphi_{it} k_{it}^{\alpha}\right)^{\frac{\xi-1}{\xi}} l_{it}^{\frac{\upsilon\xi-\upsilon-\xi}{\xi}}$$

So that we can write the **optimal labour supply** as a function of capital stock and productivity:

$$\ell(\varphi_{it}, k_{it}) = \left(\left(\frac{\omega_t}{\upsilon} \frac{\xi}{\xi - 1}\right)^{\xi} Y_t^{-1} \left(A_t \varphi_{it} k_{it}^{\alpha}\right)^{1-\xi}\right)^{\frac{1}{\upsilon\xi-\upsilon-\xi}}$$

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Optimal level of capital

An **unconstrained** intermediate goods firm chooses capital k_{it+1} according to:

$$\max_{k_{it+1}} \quad -k_{it+1} + \beta \mathbb{E}_t \left[(\pi(k_{it+1}, \varphi_{it+1}) + (1-\delta) k_{it+1} | \varphi_{it} \right]$$

So the optimal amount of capital $k^*(\varphi_{it})$ solves the following equation

$$\beta \mathbb{E}_t \left[\frac{d\pi(\varphi_{it+1}, k_{it+1})}{dk_{it+1}} | \varphi_{it} \right] = 1 + \beta\delta - \beta$$

What is $\frac{d\pi}{dk_i}$? With optimal labour demand $\ell(\cdot, \cdot)$ we can write profit as:

$$\pi(\varphi_{it}, k_{it}) = Y_t^{1/\xi} \left(A_t \varphi_{it} k_{it}^{\alpha} \ell(\varphi_{it}, k_{it})^{\upsilon} \right)^{\frac{\xi-1}{\xi}} - \omega_t \ell(\varphi_{it}, k_{it})$$

Back

Optimal level of capital

So:

$$\frac{d\pi(\varphi_{it}, k_{it})}{dk_{it}} = \frac{\xi - 1}{\xi} Y_t^{1/\xi} \left(A_t \varphi_{it} k_{it}^{\alpha} \ell(\varphi_{it}, k_{it})^{\upsilon}\right)^{-\frac{1}{\xi}}$$
$$\left(\alpha A_t \varphi_{it} k_{it}^{\alpha-1} \ell(\varphi_{it}, k_{it})^{\upsilon} + \upsilon A_t \varphi_{it} k_{it}^{\alpha} \ell(\varphi_{it}, k_{it})^{\upsilon-1} \frac{\partial \ell(\varphi_{it}, k_{it})}{\partial k_{it}}\right)$$
$$- \omega_t \frac{\partial \ell(\varphi_{it}, k_{it})}{\partial k_{it}}$$

Note that:

$$\frac{\partial \ell(\varphi_{it}, k_{it})}{\partial k_{it}} = \left(\left(\frac{\omega_t}{\upsilon} \frac{\xi}{\xi - 1}\right)^{\xi} Y_t^{-1} \left(A_t \varphi_{it}\right)^{1-\xi}\right)^{\frac{1}{\upsilon\xi - \upsilon - \xi}} \frac{\alpha(1-\xi)}{\upsilon\xi - \upsilon - \xi} k_{it}^{\frac{\alpha(1-\xi)}{\upsilon\xi - \upsilon - \xi} - 1}$$
$$= \ell(\varphi_{it}, k_{it}) \frac{\alpha(1-\xi)}{\upsilon\xi - \upsilon - \xi} k_{it}^{-1}$$

Model - Firm specific productivity

Use the following stochastic process:

$$\ln \varphi_{it} = \theta_i + w_{it}$$

where

$$\theta_i \sim \mathcal{N}\left(\mu_\theta, \sigma_\theta^2\right)$$
$$w_{i,t} = \rho_w w_{i,t-1} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim \mathcal{N}\left(0, \sigma_\varepsilon^2\right),$$

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Model - Entry and exit

- Fixed measure of potential entrants π_d
- Enter with zero debt and average capital of $\mu_{ke}\bar{k}_t^*$
- Draw productivity from stationary distribution of φ_t
- Entry takes place at the end of the period, start operating in the next period, given (x_0, φ_0)

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Calibration

Parameter	Description	Value	
Externally calibrated			
β	Discount factor	0.96	
α	Returns on capital	0.30	
η	Returns on labor	0.60	
δ	Depreciation rate	0.065	
ψ	Labour preference	2.15	
π_d	Exogenous probability of exit	0.05	
μ_{ke}	Relative size of entrants	0.10	
μ_θ	Average of permanent productivity	0	
μ_w	Average of transitory shock	0	
Internally calibrated		two shocks	one shock
ξ	Collateral constraint	0.47	0.39
σ_θ	Standard deviation of permanent productivity	0.44	NA
ρ_w	Persistence of transitory shock	0.62	0.99
σ_w	Standard deviation of transitory shock	0.18	0.5
σ_{ke}	Standard deviation of entrants	0.97	1.07

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